Geographic Information Systems- GIS 3072c

1 Overview

This course introduces geographic information systems to Geomatics and natural resources students. The course aims to provide both theoretical background and diverse practical skills needed in many applications. Students learn basic GIS data modeling and management concepts, spatial references, and analysis tools. Real world case studies involving data modeling, overlay analysis, and surface modeling are presented.

- 3 Credits - Fall Semester
- Format: 100% online
- http://elearning.ufl.edu/

Course Prerequisites: none

Instructor: Dr. Amr Abd-Elrahman (Phone: 813.757.2283, Email: **aamr@ufl.edu**)

- Please use gator link email (aamr@ufl.edu) for fastest response.
- Office hours on **Tuesday 2:00-3:00pm, Thursday 3:00-4:00pm**, and by appointment.
- Expected heavy and active **participation in the Assignments' online discussions** on Canvas.

Textbook(s) and/or readings:

Required Textbook (lab exercises):

Gorr, Wilpen L. and Kristen S. Kurland, "GIS Tutorial for ArcGIS Pro 3.1," ESRI Press, ISBN: 9781589487390. ****This specific version of the book (3.1) is required to complete course assignments.****

Recommended Textbook

Paul Bolstad (2016). GIS Fundamentals (5th edition). Eider Press. ISBN: 978-1506695877.
Note: The older or newer version of the book will work, too.

Additional Materials:

- Reading and multimedia material will be provided throughout the semester. Web links to GIS topics and data source material will be provided.

2 Learning Outcomes

At the end of this course, each student will be able to:

- Identify the concept of geographic information systems and data sources
- Utilize different national and international spatial reference systems and perform spatial reference transformation
- Model spatial and non-spatial data in relational and object-relational databases.
- Apply vector and raster data analysis and solve spatial problems using vector analysis tools
- Implement ArcGIS software in analyzing GIS data

3 Course Logistics

Students may access lectures, assignments, readings, and supporting materials through the course Canvas site as they become available.

Learning modules consisting of a lecture, readings, supporting material, and quizzes provided online for each topic. Learning modules are built on previous modules so you should complete the learning modules in the order presented.

Technology Requirements:

- A computer or mobile device with high-speed internet connection.
- A webcam, headset and/or microphone, and speakers.
- Latest version of web browser. Canvas supports only the two most recent versions of any given browser. What browser am I using?
- Use of proctoring software (Honorlock with required browser and extensions) is required and will be provided.

Synchronous online sessions may be recorded. By sharing your video, screen, or audio during any synchronous online class sessions, you are consenting to being recorded for the benefit of students who cannot attend live as well as for class review during the current semester. If you have special circumstances or concerns about privacy, it is your responsibility to discuss it with your instructor.

ArcGIS Software Access

The primary and recommended method of accessing the ArcGIS Pro software for most students is by downloading the software and installing it locally on your computer. Students whose computers are running operating systems other than Windows (e.g. Apple Mac OS), or who don't want to install the software locally on their machines, will use the software through UF APPS (**https://info.apps.ufl.edu/**). Be aware that if you use UFAPPS, you will not be able to store data locally to your computer and will need to follow the provided detailed instructions on how to access the UFAPPS storage folder. These instructions will be provided early in the semester. Please watch the Lab recordings and review the material in Module 1-week 1 page of the course website in canvas.

Using Zoom Software

Review and office hour meetings will be conducted using the Zoom software. The software is accessed by clicking a link posted by the instructor through e-Learning or email. The instructor will schedule the sessions and provide the link to you.

Please check https://support.zoom.us/hc/en-us/articles/201362023-System-Requirements-for-PC-Mac-and-Linux for more information about Zoom software use requirement. Zoom supports almost any operating system including Windows, Macintosh, and Linux, as well as the most widely used browsers including Edge, Firefox, Safari, and Chrome. A microphone is also needed to communicate with the instructor and the students attending the session.

In Class Recording

Students are allowed to record video or audio of class lectures. However, the purposes for which these recordings may be used are strictly controlled. The only allowable purposes are (1) for personal educational use, (2) in connection with a complaint to the university, or (3) as evidence in, or in preparation for, a criminal or civil proceeding. All other purposes are prohibited.

Specifically, students may not publish recorded lectures without the written consent of the instructor. A "class lecture" is an educational presentation intended to inform or teach enrolled students about a particular subject, including any instructor-led discussions that form part of the presentation, and delivered by any instructor hired or appointed by the University, or by a guest instructor, as part of a University of Florida course. A class lecture does not include lab sessions, student presentations, clinical presentations such as patient history, academic exercises involving solely student participation, assessments (quizzes, tests, exams), field trips, private conversations between students in the class or between a student and the faculty or lecturer during a class session.

Publication without permission from the instructor is prohibited. To "publish" means to share, transmit, circulate, distribute, or provide access to a recording, regardless of format or medium, to another person (or persons), including but not limited to another student within the same class section. Additionally, a recording, or transcript of a recording, is considered published if it is posted on or uploaded to, in whole or in part, any media platform, including but not limited to social media, book, magazine, newspaper, leaflet, or third-party note/tutoring services. A student who publishes a recording without written consent may be subject to a civil cause of action instituted by a person injured by the publication and/or discipline under UF Regulation 4.040 Student Honor Code and Student Conduct Code.

3.1 Assignments & Deliverables

All meeting times and due date times are provided in Eastern time. Students located in a different time zone should be aware of their Canvas time zone settings to ensure they know when due dates are and when assignments close.

<u>Tutorials</u>

The ArcGIS Tutorials for ArcGIS Pro 3.1 book is used in this class. The book is divided into chapters. Each chapter has a specific set of objectives and is divided into tutorials. You are required to do chapters and (tutorials) as assigned throughout the semester according to the assignments released in the course website. **A summary lab worksheet on each chapter that includes snapshots of specific screens in each tutorial and a brief summary of what you learned in the chapter should be submitted according to the due date specified in canvas. Collaboration on software issues is allowed; however the tutorials must be performed by each student independently.**

PLEASE MAKE SURE THAT YOUR SCREEN CAPTURES INCLUDE THE **COMPUTER DATE AND TIME** AT THE LOWER RIGHT CORNER OF THE SCREEN TO RECEIVE CREDIT. The screen snapshots can be taken using any of the online freeware available for this purpose, the windows snipping tool, or using the ctrl-PrintScr (or Fn-PrintScr) to capture and ctrl-v to paste the snapshot.

Assignment feedback will be communicated through the canvas course website. Comments will be provided mainly using the grading portal of each assignment. Students are encouraged to review and digest the comments promptly to avoid recurring errors in future assignments.

Participation

Virtual (online) discussion topics are included in the course website (Canvas). You are strongly encouraged to read, post and interact in these discussions. Please contribute positively to the discussions by providing useful/tested technical tips as well as innovative and critical thoughts. You are also encouraged to enrich course resources with online material. Participation grades will be issued based on the quantity and quality of your participation in the online course discussion, including some discussion board topics marked for participation grade and require your participation.

Quizzes

Quizzes will be delivered via the Canvas course site throughout the semester. These short quizzes will assess the student's understanding of lecture material, readings, and laboratory assignments.

Final Exam

The final exam will be conducted online using the course e-learning (canvas) website and Honorlock proctoring software through canvas on **12/03/2025 from 5:30-7:30pm**. The exam must be taken during the exam window **and are not open for an entire day. Exam time is specified in Eastern time.** It is expected that you make accommodations to be available during the exam time. **Exam conflicts must be communicated with the professor and arrangements made by 11/19. Exams are an individual assessment and collaboration is strictly prohibited.**

3.2 Grades & Grading Scale

Grading Item	Grade Percentage	Description
Tutorials	**45%**	This includes performing assigned tutorials from 'GIS Tutorial for ArcGIS Pro 3.1' book and submitting lab worksheets. Please see the 'Tutorials' section for more details.
Participation	**10%**	Class participation includes class attendance and participation in online discussions. Please see the 'Class Participation' section for more details.
Quizzes	**20%**	Online quiz covering lab and course material.
Final Exam	**25%**	The final exam is scheduled for 12/03/2025 from 5:30-7:30pm.

Please note that we are using the + and - grading scale encouraged by UF. For more information about the new grading system, please visit http://www.isis.ufl.edu/minusgrades.html.

Be aware that the calculated canvas overall/final grade may not be accurate due to having different sections with different grading scales in the same canvas course, which does not allow implementing multiple grading schemes. Use the above grading scale combined with the individual item scores in canvas to calculate your overall grade.

Grade Scale: For information on current UF policies for assigning grade points, see https://catalog.ufl.edu/ugrad/current/regulations/info/grades.aspx

Letter Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E
Corresponding Course Score	95-100	90-94	85-89	80-84	75-79	70-74	65-69	60-64	55-59	50-54	45-49	0-44
Grade Points	4	3.67	3.33	3	2.67	2.33	2	1.67	1.33	1	0.67	0

3.3 Late Submissions & Make-up Requests

It is the responsibility of the student to access on-line lectures, readings, quizzes, and exams and to maintain satisfactory progress in the course.

Tutorial worksheets turned in after the due date will have points deducted. To receive points for a late assignment, the report must be turned in no later than two weeks past the due date. Each day the assignment is late (each day being a range from 1 minute to 24 hours) will result in a 3% late deduction in possible total points (for example, perfect work one day late would receive a 97%, and a 70% grade would receive a 67% if submitted one day late). A maximum of 42% will be deducted before the submission window closes 2 weeks after the deadline. Lab reports and projects will not be accepted after two weeks from the deadline. **Quizzes cannot be submitted late – be aware that the close date on canvas is the same as the due date.**

Examples for the reasons justifying missing class activities can be found in https://catalog.ufl.edu/ugrad/current/regulations/info/attendance.aspx. Please contact us immediately (within 24 hours) if you have any unusual circumstances to arrange for make-up plans.

Computer or other hardware failures, except failure of the UF e-Learning system, will not excuse students for missing assignments. Any late submissions due to technical issues MUST be accompanied by the ticket number received from the Helpdesk when the problem was reported to them. The ticket number will document the time and date of the problem. You MUST e-mail your instructor within 24 hours of the technical difficulty if you wish to request consideration.

For computer, software compatibility, or access problems call the HELP DESK phone number—352-392-HELP = 352- 392-4357 (option 2).

Requirements for class attendance and make-up exams, assignments, and other work are consistent with university policies that can be found at: https://catalog.ufl.edu/ugrad/current/regulations/info/attendance.aspx.

4 Course Content

Learning Modules, Lecture, Tutorial, and Quiz Schedule[#*]

Week Of (Week #)	Module	Lecture Topic	Reading (GIS Fundamentals Book – based on 5th Edition)	Tentative: Lab Topic (GIS Tutorial 1 for ArcGIS Pro Book) and Quizzes
Aug. 18 (1)	1. Introduction to GIS and Data Sources	Course Outlines & GIS Introduction		Lab Setup Assignment: Software & Data Access
Aug. 25 (2)		Introduction to ArcGIS Software & Digital Data Sources	Ch 1 pp. 1-21; Ch 2 pp. 29-53	GIS Tutorial 1 For ArcGIS Pro: Chapter 1
Sept. 1 (3)		Data Sources Continued (Global Navigation Satellite Systems)	Ch 5 pp. 203-221; Ch 7 pp. 299-327	GIS Tutorial 1 For ArcGIS Pro: Chapter 2
Sept. 08 (4)		Data Sources Continued (Aerial & Satellite Images)	Ch 6 pp. 249 – 291	Quiz 1
Sept. 15 (5)		Data Sources Continued (Lidar) & Spatial Reference		GIS Tutorial 1 For ArcGIS Pro: Chapter 3
Sept. 22 (6)	2. Spatial References	Horizontal Datum & Map Projections	Ch 3 pp. 85-137	Quiz 2
Sep 29 (7)	3. Data Modeling and Management	Data Modeling & Management (Conceptual and Logical)	Ch 8 pp. 331 – 349	GIS Tutorial 1 For ArcGIS Pro: Chapter 4
Oct. 6 (8)		Data Modeling & Management Continued (Logical Modeling DBMS)	Ch 8 pp. 350 –364	Quiz 3
Oct. 13 (9)		Data Modeling (Physical Implementation Demo)		GIS Tutorial 1 For ArcGIS Pro: Chapters 5 & 7
Oct. 20 (10)	4. GIS Vector Analysis	Vector Analysis Introduction	Ch 9 pp. 373 - 419	Quiz 4
Oct. 27 (11)		Vector Analysis Continued	Ch 9 pp. 420 - 428	GIS Tutorial 1 For ArcGIS Pro: Chapter 6
Nov. 3 (12)	5. Surface Modeling and Raster Analysis	Surface Modeling	Ch 12 pp. 519-533	Quiz 5
Nov. 10 (13)		Raster Analysis	Ch 10 pp. 443 – 473	GIS Tutorial 1 For ArcGIS Pro: Chapter 9
Nov. 17 (14)		Raster Analysis Continued		Quiz 6
Nov. 24 (15)		Thanksgiving Holiday		
Dec. 1 (16)		ArcGIS Online & Exam Review		**Final Exam**: 12/03 5:30-7:30pm EST

[#]Schedule is tentative and subject to change due to actual course delivery circumstances
*Accurate release and due dates for each activity (e.g. assignment, quiz, and exam) are in Canvas.

5 Policies and Requirements

This course plan and syllabus are subject to change in response to student and instructor needs. Any changes will be clearly communicated in advance through Canvas.

5.1 Communication Courtesy and Professionalism

Just as in any professional environment, meaningful and constructive dialogue is expected in this class and requires a degree of mutual respect, willingness to listen, and tolerance of opposing points of view. **Respect for individual differences and alternative viewpoints will be maintained in this class at all times.** All members of the class are expected to follow rules of common courtesy, decency, and civility in all interactions. Failure to do so will not be tolerated and may result in loss of participation points and/or referral to the Dean of Students' Office.

5.2 Semester Evaluation Process

Student assessment of instruction is an important part of efforts to improve teaching and learning.

At approximately the mid-point of the semester, the School of Forest Resources & Conservation will request anonymous feedback on student satisfaction on various aspects of this course. These surveys will be sent out through Canvas and are not required but encouraged. This is not the UF Faculty Evaluation!

At the end of the semester, students are expected to provide professional and respectful feedback on the quality of instruction in this course by completing course evaluations online via GatorEvals. Guidance on how to give feedback in a professional and respectful manner is available at https://gatorevals.aa.ufl.edu/students/. Students will be notified when the evaluation period opens, and can complete evaluations through the email they receive from GatorEvals, in their Canvas course menu under GatorEvals, or via https://ufl.bluera.com/ufl/. Summaries of course evaluation results are available to students at https://gatorevals.aa.ufl.edu/public-results/.

5.3 UF Honor Code and Academic Honesty Policy

UF students are bound by The Honor Pledge which states "We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honor and integrity by abiding by the Honor Code. On all work submitted for credit by students at the University of Florida, the following pledge is either required or implied: "On my honor, I have neither given nor received unauthorized aid in doing this assignment." The Conduct Code specifies a number of behaviors that are in violation of this code and the possible sanctions. See the UF Conduct Code website for more information. If you have any questions or concerns, please consult with the instructor or TAs in this class.

It is assumed that you will complete all work independently in each course unless the instructor provides explicit permission for you to collaborate on course tasks (e.g. assignments, papers, quizzes, exams). It is your individual responsibility to know and comply with all university policies and procedures regarding

academic integrity and the Student Honor Code. Violations of the Honor Code at the University of Florida will not be tolerated. Violations will be reported to the Dean of Students Office for consideration of disciplinary action. For more information regarding the Student Honor Code, please see: https://policy.ufl.edu/regulation/4-040/

5.4 Privacy Statement

Our class sessions may be audio visually recorded for students in the class to refer back and for enrolled students who are unable to attend live. Students who participate with their camera engaged or utilize a profile image are agreeing to have their video or image recorded. If you are unwilling to consent to have your profile or video image recorded, be sure to keep your camera off and do not use a profile image. Likewise, students who un-mute during class and participate orally are agreeing to have their voices recorded. If you are not willing to consent to have your voice recorded during class, you will need to keep your mute button activated and communicate exclusively using the "chat" feature, which allows students to type questions and comments live. The chat will not be recorded or shared. As in all courses, unauthorized recording and unauthorized sharing of recorded materials is prohibited.

5.5 In-class recordings

Students are allowed to record video or audio of class lectures. However, the purposes for which these recordings may be used are strictly controlled. The only allowable purposes are (1) for personal education use, (2) in connection with a complaint to the university, or (3) as evidence in, or in preparation for, a criminal or civil proceeding. All other purposes are prohibited. Specifically, students may not publish recorded lectures without the written consent of the instructor.

A “class lecture” is an educational presentation intended to inform or teach enrolled students about a particular subject, including any instructor-led discussions that form part of the presentation, and deliver by an instructor hired or appointed by the University, or by a guest instructor, as part of a University of Florida course. A class lecture does not include lab sessions, student presentations, clinical presentation such as patient history, academic exercises involving solely student participation, assessments (quizzes, tests, exams), field trips, private conversations between students in the class or between a student and the faculty or guest lecturer during a class session.

5.6 Services for Students with Disabilities:

Students with disabilities who experience learning barriers and would like to request academic accommodations should connect with the Disability Resource Center. See the “Get Started With the DRC” webpage on the Disability Resource Center site https://disability.ufl.edu/get-started/. It is important for students to share their accommodation letter with their instructor and discuss their access needs, as early as possible in the semester.

5.7 Software Use

All faculty, staff and students of the university are required and expected to obey the laws and legal agreements governing software use. Failure to do so can lead to monetary damages and/or criminal

penalties for the individual violator. Because such violations are also against university policies and rules, disciplinary action will be taken as appropriate.

6 Campus Helping Resources

6.1 Whole Gator App

The Whole Gator and website and app connects UF students with resources dedicated to supporting overall health and well-being. In addition to many of the resources below it also has strategies to practice self-care. https://one.uf.edu/whole-gator/topics

6.2 Health and Wellness

- U Matter, We Care: If you or someone you know is in distress, please contact umatter@ufl.edu, 352-392-1575, or visit U Matter, We Care website to refer or report a concern and a team member will reach out to the student in distress.
- Counseling and Wellness Center: Visit the Counseling and Wellness Center website or call 352-392-1575 for information on crisis services as well as non- crisis services.
- Student Health Care Center: Call 352-392-1161 for 24/7 information to help you find the care you need, or visit the Student Health Care Center website.
- University Police Department: Visit UF Police Department website or call 352-392-1111 (or 9-1-1 for emergencies).
- UF Health Shands Emergency Room / Trauma Center: For immediate medical care call 352-733-0111 or go to the emergency room at 1515 SW Archer Road, Gainesville, FL 32608; Visit the UF Health Emergency Room and Trauma Center
website.
- GatorWell Health Promotion Services: For prevention services focused on optimal wellbeing, including Wellness Coaching for Academic Success, visit the GatorWell website or call 352-273-4450.
- Student Success Initiative, http://studentsuccess.ufl.edu

6.3 Academic Resources

- E-learning technical support: Contact the UF Computing Help Desk at 352-392-4357 or via e-mail at helpdesk@ufl.edu.
- Career Connections Center: Reitz Union Suite 1300, 352-392- 1601. Career assistance and counseling services.
- Library Support: Various ways to receive assistance with respect to using the libraries or finding resources. Call 866-281-6309 or email ask@ufl.libanswers.com for more information.
- Teaching Center: 1317 Turlington Hall, 352-392-2010 or to make an appointment 352- 392-6420. General study skills and tutoring.
- Writing Studio: Daytime (9:30am-3:30pm): 2215 Turlington Hall, 352-846-1138 | Evening (5:00pm-7:00pm): 1545 W University Avenue (Library West, Rm. 339). Help brainstorming, formatting, and writing papers.

6.4 Student Complaints

- Academic Complaints: Office of the Ombuds; Visit the Complaint Portal webpage for more information.
- Enrollment Management Complaints (Registrar, Financial Aid, Admissions): View the Student Complaint Procedure webpage for more information.
- Residential Course: https://www.ombuds.ufl.edu/complaint-portal/
- Online Course: https://pfs.tnt.aa.ufl.edu/state-authorization-status/#student-complaint